NES AI FINANCE & INSURANCE INTELLIGENCE PLATFORM

Error Handling & Diagnostics Plan

Document ID: NES-OPS-015 | **Version:** 1.0

Build approach: Cline-assisted development • Local-first Windows application

Scope: Personal research, paper trading and insurance document assistance

Failure states, logging and local troubleshooting

1. Objectives

- Make failures understandable without exposing internals or secrets.
- Distinguish user errors, provider failures, stale data and application bugs.
- Provide enough diagnostic context to troubleshoot locally.

2. Error categories

Category	User-facing behaviour
Validation	Explain the field to fix; preserve safe entered values.
Provider unavailable	Show connection status, last update and retry option.
Stale/partial data	Warn that data is stale or incomplete; do not imply live status.
Document parse failure	Explain file/page failure and allow another file or manual entry.
Local model unavailable	Offer retry or non-AI workflow where possible.
Unexpected error	Show safe message and local diagnostic reference; no stack trace.
Cancelled operation	Mark cancelled rather than failed or successful.

3. Logging rules

- Use structured logs with timestamp, level, request/operation ID and safe status fields.
- Never log API keys, passwords, full policy text or unnecessary personal identifiers.
- Redact file paths and sensitive metadata where appropriate.
- Rotate or limit log size and document log retention.

4. Diagnostics screen

- Application version and runtime status.
- Database/local storage status where implemented.
- Market provider connection status and last successful update.
- Local AI model availability and resource errors.
- Recent safe error summaries with retry or documentation link.

5. Acceptance

- Every asynchronous workflow has a meaningful failure state.
- Retry does not duplicate destructive or financial operations.
- Errors can be diagnosed without collecting unnecessary sensitive data.